NPC CAPSTONE PROJECT

Malware Analysis Sandbox Using VirtualBox

Technical Report & Evidence Dossier

Student	Aditya Devade
Program	B.Tech — Computer Science & Engineering (Cybersecurity)
Lab Platform	Oracle VirtualBox / Kali Linux
Report date	19 October 2026
Scope	Isolated environment setup, benign system observation, local TCP diagnostic, and VirusTotal EICAR re

Prepared from the supplied lab screenshots and user-reported configuration. The report distinguishes completed observations from uncompleted assignment items.

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1. Executive summary

This report records work performed for the NPC capstone assignment on a VirtualBox-based malware-analysis sandbox. The evidence shows a Kali Linux virtual machine with VirtualBox network adapters reported disabled, shared folders reported empty, and a clean baseline snapshot created. A benign Bash observation script was executed to print basic host, process, operating-system, and interface details. The terminal output showed loopback and a Docker bridge interface in the DOWN state, and no default route was displayed.

A local TCP listener/connection diagnostic was also attempted, and VirusTotal pages were reviewed for the EICAR antivirus test file. These are useful supporting observations, but they do not constitute execution of a malware sample, a functioning reverse/bind shell, or a full Wireshark traffic analysis. Those limitations are explicitly documented rather than overstated.

2. Objectives and scope

The intended assignment concerns creating an isolated VM sandbox, establishing a clean snapshot, observing system and network changes, reviewing sample reputation when available, and restoring the VM to a known-good state.

Objectives addressed by the available evidence

• Document the reported VirtualBox isolation configuration.

• Capture Kali Linux interface and route information.

• Run a harmless observation script and record process/system output.

• Record a SHA-256 hash reported by the Kali terminal.

• Inspect a local TCP listener and connection diagnostic.

• Review VirusTotal Details and Relations information for the EICAR test file.

Scope boundary

No actual malicious executable or script was supplied or executed as part of the evidence shown here. The sandbox script only gathers system information and appends output to a log. The VirusTotal artifact is EICAR, a benign antivirus test file. This report therefore describes a sandbox setup and benign monitoring exercise, not a completed malware-behavior analysis.

3. Lab environment and isolation

Platform: Oracle VirtualBox with a Kali Linux guest. The user reported that Adapter 1, Adapter 2, Adapter 3, and Adapter 4 were all disabled, and that the Shared Folders configuration was empty. The guest terminal screenshot showed the loopback interface (`lo`) with 127.0.0.1/8 and ::1/128, plus `docker0` at 172.17.0.1/16 in the DOWN state. The route table displayed only the Docker bridge network route with linkdown; no default route was visible.

Interpretation: the displayed network state is consistent with a guest that had no active external network path at capture time. It is not, by itself, a complete proof that every VirtualBox integration feature was disabled. For repeatability, the final submission should include the VirtualBox Network and Shared Folders settings screenshots as separate evidence.

Control	Reported / observed state	Assessment
VirtualBox adapters 1–4	Disabled (user-confirmed)	Supports isolation
Shared Folders	Empty (user-confirmed)	Reduces host/guest file sharing
Guest interfaces	lo active; docker0 DOWN	No active non-loopback interface shown
Default route	Not shown in `ip route` output	No default route observed

4. Methodology

4.1 Baseline snapshot

A VirtualBox snapshot named `Clean-Baseline` was created while the VM was powered off. The screenshot history later shows additional snapshots named `After-Observation` / similar, but the exact final restore outcome is not conclusively documented in the supplied evidence.

4.2 Network-state inspection

The commands `ip -br address` and `ip route` were used to review interfaces and routing. The output is included in the terminal evidence.

4.3 Benign system observation

A Bash script named `sandbox_demo.sh` printed the date, PID/PPID, current user, hostname, `uname -a`, selected process information from `ps`, and a brief interface summary from `ip -br address`. It appended output to `~/npc-capstone/logs/experiment.log`.

4.4 Script integrity

The terminal showed a SHA-256 hash for the script: `81cf4d1341fee6d511ba925155dd3e85d028c1a53ee8a26ab83221690c173d8c`. The report records this as the user-captured value.

4.5 TCP diagnostic

The evidence shows use of `ss -lnt` and `sudo ss -lntp`, which displayed a listening socket at 127.0.0.1:44263 owned by `containerd`. A prior local `nc` attempt showed a connection test, but the evidence does not prove that a shell was provided over the connection.

4.6 VirusTotal reference lookup

The VirusTotal Details screenshot identifies the artifact as an EICAR antivirus test file, 68 bytes, with a SHA-256 beginning `275a021bbfb6489e54d471899f7db9d1663fc695ec2fe2a2c4538aabf651fd0f`. Its history section shows first submission in 2006 and a last analysis date of 2026-10-19 in the screenshot.

5. Observations and results

Area	Evidence-based observation	Result
Isolation	Adapters disabled; Shared Folders empty (user-confirmed)	Configured for isolation, based on reported settings
Interfaces	lo at 127.0.0.1; docker0 DOWN at 172.17.0.1/16	No active external interface shown
Routing	Only 172.17.0.0/16 linkdown route displayed	No default route observed
Process details	Script PID 4353; PPID 2993; user kali; host kali	Script ran and collected basic host details
System details	Linux kernel string shown in terminal screenshot	OS details captured
Local TCP	Listener at 127.0.0.1:44263 attributed to containerd	Listener inspection performed; not evidence of a shell
Hash	SHA-256 recorded in terminal	Integrity identifier recorded
VirusTotal	EICAR test file; Details and Relations screenshots	Reference artifact reviewed; not the demo script

6. VirusTotal reference analysis

The provided VirusTotal Details screenshot lists the file name `eicar.com`, size 68 bytes, and file magic/type identifying it as an EICAR antivirus test file. The page also shows a first submission date of 2006-05-22 and a last analysis date of 2026-10-19. The screenshot's vendor results include numerous EICAR/test-file labels. This is expected for a standardized antivirus test string and should not be interpreted as proof that the Kali observation script is malicious.

The Relations screenshot shows lists of contacted domains, contacted IP addresses, execution parents, PE resource parents, dropped files, and a graph summary. These are VirusTotal relationship records associated with the EICAR artifact and its broader dataset context; they are not evidence that those domains, IPs, parent files, or dropped files were contacted or created by the user's Kali VM during this exercise.

No VirusTotal scan of `sandbox_demo.sh` is documented in the supplied screenshots. The EICAR record is included only as an example of interpreting a known antivirus test artifact. First-seen and historical data refer to the artifact record, not the date this lab script was created.

7. Snapshot and recovery status

A VirtualBox snapshot named `Clean-Baseline` was created before the observation activity. The user later attempted to restore it and created additional snapshot entries. The screenshots supplied here do not conclusively show a completed post-restore verification against the baseline. Therefore, this report records the baseline snapshot as created, but does not claim that full recovery was verified.

To finish this section, restore `Clean-Baseline` only when ready to discard the current guest state, and verify the guest returns to its expected clean state. Because the VM may contain other work, make a separate backup/snapshot of the current state first and ensure important files are stored outside the VM before restoring.

8. Security interpretation

The main security control demonstrated is reducing network exposure before performing lab observations. Disabling all virtual network adapters and clearing shared folders helps limit communication between the guest, host, and external networks. A snapshot provides a rollback point. The benign script offers a repeatable way to collect basic state, but it does not simulate malicious behavior. Process and socket observations must be interpreted with context: the `containerd` listener shown at 127.0.0.1:44263 is a local service and should not be mislabeled as malware based only on its presence.

9. Limitations and remaining work

• No actual malware sample was executed; the script is benign.

• A functional reverse shell or bind shell was not demonstrated. The local TCP check is only a connection/listener diagnostic.

• No Wireshark capture or packet-level analysis is visible in the supplied evidence.

• The VirusTotal lookup concerns EICAR, not the sandbox script or a real malware specimen.

• A post-restore verification screenshot and comparison with the baseline are not included in the uploaded evidence.

• The submitted screenshots do not show the VirtualBox adapter settings page; the disabled state is recorded from the user's confirmation.

These are not cosmetic omissions: if the instructor requires every original assignment objective, these items should be completed in the isolated lab before representing the assessment as fully complete.

10. Conclusion

The available evidence supports that a Kali Linux VM was prepared with reported network isolation controls, a baseline snapshot was created, and a benign script successfully collected process, OS, and network-interface information. The script's SHA-256 identifier was recorded. A local TCP socket diagnostic and a VirusTotal reference review of the harmless EICAR test file were also documented. The report deliberately distinguishes these completed activities from full malware execution analysis, reverse/bind-shell testing, Wireshark analysis, and confirmed post-restore validation, which are not established by the supplied evidence.

Appendix A — Evidence screenshots

Figure A1. Kali network state and benign observation script output.



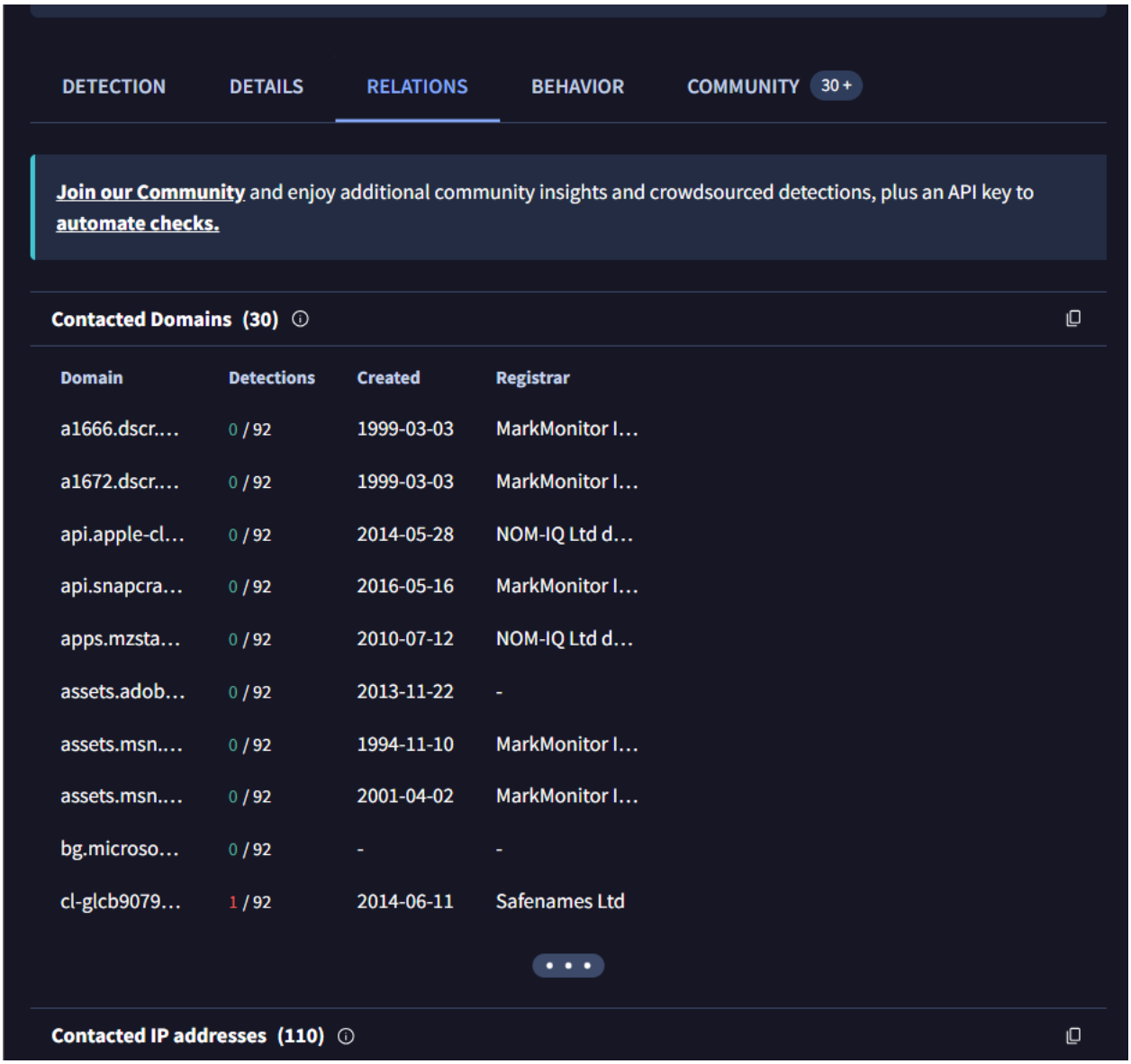
Shows loopback and a DOWN Docker bridge, the route output, and script output including PID, user, hostname, kernel details, process information, and interface summary.

Figure A2. Duplicate capture of the Kali network and script output.



This appears to duplicate the same terminal evidence as Figure A1; retained for traceability but one copy is sufficient for final submission.

Figure A3. VirusTotal Relations tab for the EICAR test artifact.



Shows relationship categories such as contacted domains/IPs and file relationships. These are artifact-level relations, not proof of activity from this lab VM.

Figure A4. Local TCP listener inspection.



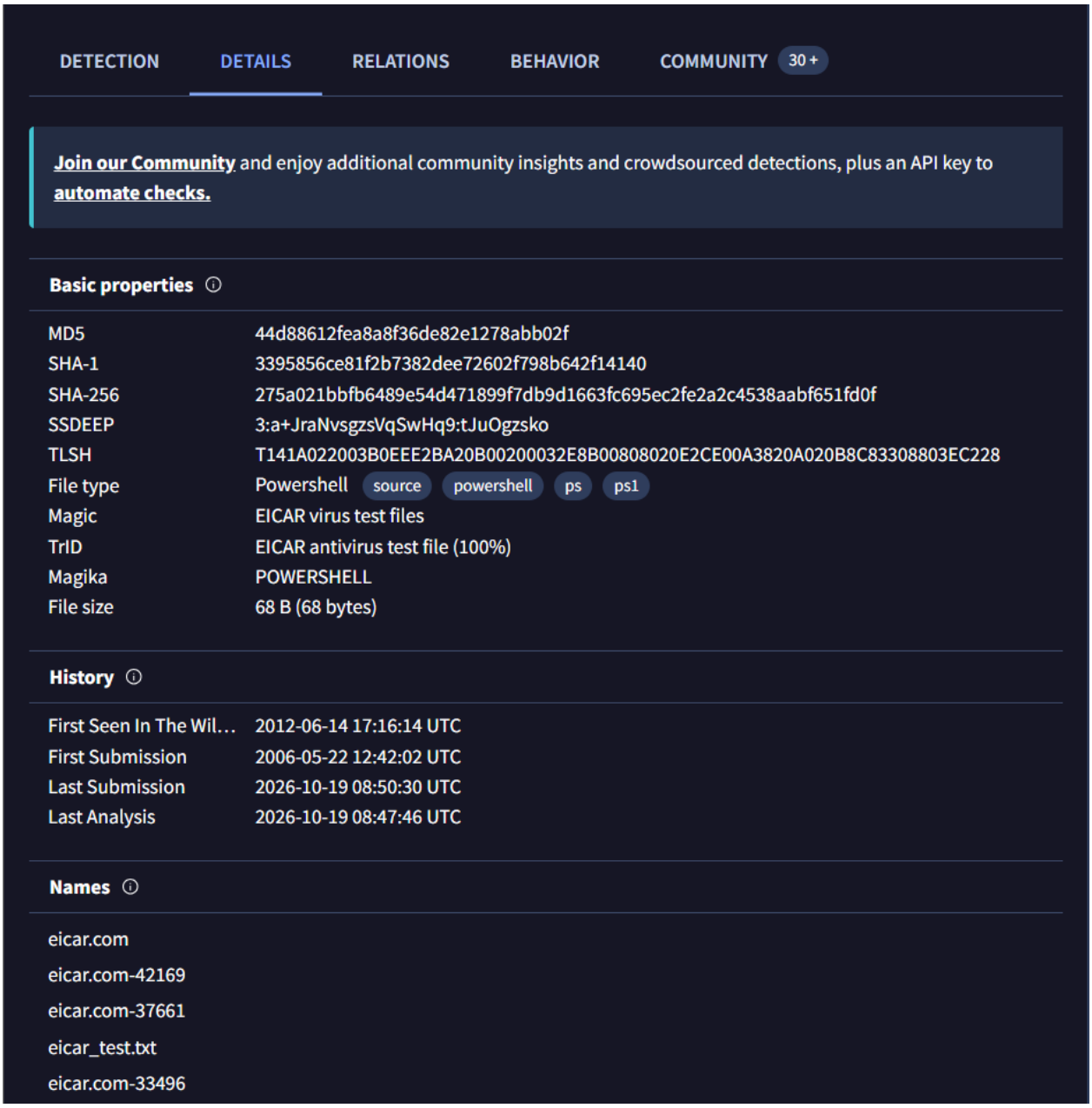
The `ss` output shows 127.0.0.1:44263 in LISTEN state and identifies `containerd` as the owning process.

Figure A5. Source code of the benign observation script.



The script records date, process identifiers, user, hostname, kernel details, selected process information, and network interface summary.

Figure A6. VirusTotal Details for the EICAR antivirus test file.



Displays file properties, SHA-256, 68-byte size, and historical submission/analysis dates. EICAR is a harmless antivirus test artifact.

Appendix B — Script and integrity notes

Script file included in the project: `scripts/sandbox_demo.sh`. It is intentionally benign and does not open a network listener, execute commands received over a socket, modify system configuration, or attempt persistence.

SHA-256 recorded in the user's Kali terminal: `81cf4d1341fee6d511ba925155dd3e85d028c1a53ee8a26ab83221690c173d8c`.

The source was reconstructed from the screenshot for inclusion in this package. Since a screenshot transcription can differ from the exact original bytes, the locally calculated hash of the reconstructed copy is not substituted for the terminal-recorded hash above.

Evidence integrity note: screenshots were supplied by the user in the current conversation. The report includes those screenshots and captions; no screenshot was fabricated.